IT 340: BUSINESS INTELLIGENCE SYSTEMS

5.4 Project 1 – Baseball Dataset, Part 5

Using Advanced SQL

Submitted By:

Chinenye Edith Ugwu

Course Code: IT 340

Course Name: Business Intelligence Systems

Lecturer:

Brother Mann

September 2026

Executive Summary

Which Schools Produce the Best Baseball Players?

Introduction

Management asked which schools produce the best players, with “best” defined as reaching at least 30 runs batted in (RBI). An RBI is credited to a batter when his at-bat allows a teammate to score, so it is a standard measure of run production. This analysis used the Lahman Baseball Database, which links each player’s major league batting record to the colleges he attended.

A player was counted as successful if he recorded 30 or more RBI in at least one major league season. Each player was counted once per school, however many years he spent there, so a school is credited for the number of different successful players it sent to the major leagues rather than for one star’s long career. Two SQL queries were used. The first ranks schools, and the second identifies the top run producer from each of the leading schools.

Query 1: Ranking the Schools

```
WITH clean_batting AS (
    SELECT
        playerid,
        yearid,
        COALESCE(NULLIF(rbi, '')::numeric, 0) AS rbi
    FROM laman.batting
),
player_season AS (
    SELECT playerid, yearid, SUM(rbi) AS season_rbi
    FROM clean_batting
    GROUP BY playerid, yearid
),
player_summary AS (
    SELECT
        playerid,
        COUNT(*) FILTER (WHERE season_rbi >= 30) AS seasons_30plus
    FROM player_season
    GROUP BY playerid
),
school_players AS (
    SELECT DISTINCT playerid, schoolid
    FROM laman.collegeplaying
),
school_totals AS (
    SELECT
        sp.schoolid,
        COUNT(*) AS batters_from_school,
        COUNT(*) FILTER (WHERE ps.seasons_30plus > 0) AS players_with_30_rbi,
        COALESCE(SUM(ps.seasons_30plus), 0) AS total_30_rbi_seasons
    FROM school_players sp
    JOIN player_summary ps ON sp.playerid = ps.playerid
    GROUP BY sp.schoolid
)
SELECT
    RANK() OVER (ORDER BY st.players_with_30_rbi DESC) AS school_rank,
    s.c2 AS school,
    st.players_with_30_rbi,
    st.batters_from_school,
    ROUND(100.0 * st.players_with_30_rbi / st.batters_from_school, 1) AS pct_reaching_30_rbi,
    st.total_30_rbi_seasons
FROM school_totals st
```

```
JOIN laman.schools s ON st.schoolid = s.c1
ORDER BY school_rank, school
LIMIT 10;
```

How Query 1 Works

- **Cleaning and totaling.** The batting figures are tidied (blank values are treated as zero) and each player's yearly RBI are totaled, so a player who changed teams mid-season is not counted twice.
- **Finding successful players.** For each player, the query counts the seasons in which he reached 30 or more RBI.
- **Linking players to schools.** The college records are used to find which schools each player attended, with each player–school pair listed only once.
- **Summarizing each school.** For every school, the query counts the successful players, all batters the school sent to the major leagues, and the share who succeeded.
- **Ranking with a window function.** The RANK function orders schools from most to fewest successful players. Schools with the same total share the same rank, and the next rank is skipped accordingly.

Query 1 Results

The table below shows the top ten schools by number of players with at least one 30+ RBI season.

Rank	School	Players with 30+ RBI	Batters in Majors	% Reaching 30+ RBI	30+ RBI Seasons
1	Arizona State University	36	101	35.6%	229
2	University of Southern California	30	104	28.8%	188
3	University of California, Los Angeles	27	70	38.6%	166
4	Stanford University	22	86	25.6%	99
5	University of Alabama	20	60	33.3%	102
5	University of California, Berkeley	20	57	35.1%	100
5	University of Miami	20	54	37.0%	130
8	University of Michigan	18	76	23.7%	108
9	University of Texas at Austin	17	107	15.9%	96
10	Georgia Institute of Technology	16	52	30.8%	102

Arizona State University leads all schools, with 36 players reaching 30 or more RBI in a season out of 101 who reached the major leagues, and 229 such seasons in total. The University of Southern California is second with 30 players, and the University of California, Los Angeles is third with 27. Three schools

(Alabama, California–Berkeley, and Miami) tie for fifth place with 20 players each, which is why the next rank is eighth.

The percentage column adds an important point. UCLA has the best success rate at 38.6%, followed closely by Miami (37.0%) and Berkeley (35.1%), meaning that a very high share of their major league batters became solid run producers. The University of Texas at Austin, by contrast, sent 107 batters to the major leagues, but only 15.9% reached the threshold. Santa Clara University also ties Georgia Tech at 16 players for tenth place; it is not shown in this table, which is limited to ten rows, but it appears in the second table below.

Query 2: The Top Run Producer at Each School

```
WITH clean_batting AS (
    SELECT
        playerid,
        yearid,
        COALESCE(NULLIF(rbi, '')::numeric, 0) AS rbi
    FROM laman.batting
),
player_season AS (
    SELECT playerid, yearid, SUM(rbi) AS season_rbi
    FROM clean_batting
    GROUP BY playerid, yearid
),
player_summary AS (
    SELECT
        playerid,
        SUM(season_rbi) AS career_rbi,
        COUNT(*) FILTER (WHERE season_rbi >= 30) AS seasons_30plus
    FROM player_season
    GROUP BY playerid
),
school_players AS (
    SELECT DISTINCT playerid, schoolid
    FROM laman.collegeplaying
),
school_totals AS (
    SELECT
        sp.schoolid,
        COUNT(*) FILTER (WHERE ps.seasons_30plus > 0) AS players_with_30_rbi
    FROM school_players sp
    JOIN player_summary ps ON sp.playerid = ps.playerid
    GROUP BY sp.schoolid
),
top_schools AS (
    SELECT
        schoolid,
        players_with_30_rbi,
        RANK() OVER (ORDER BY players_with_30_rbi DESC) AS school_rank
    FROM school_totals
),
ranked_players AS (
    SELECT
        sp.schoolid,
        sp.playerid,
        ps.career_rbi,
        ROW_NUMBER() OVER (PARTITION BY sp.schoolid ORDER BY ps.career_rbi DESC) AS rn
    FROM school_players sp
    JOIN player_summary ps ON sp.playerid = ps.playerid
)
SELECT
    ts.school_rank,
    s.c2 AS school,
```

```
    pe.namefirst,
    pe.namelast,
    rp.career_rbi
FROM top_schools ts
JOIN ranked_players rp ON ts.schoolid = rp.schoolid AND rp.rn = 1
JOIN laman.schools s ON ts.schoolid = s.c1
JOIN laman.people pe ON rp.playerid = pe.playerid
WHERE ts.school_rank <= 10
ORDER BY ts.school_rank, school;
```

How Query 2 Works

The second query reuses the same steps to rank the schools, then adds a second window function. The ROW_NUMBER function, combined with PARTITION BY, restarts its count for each school and orders that school's players by career RBI. The player numbered one is the school's top run producer. The final step keeps only the ten leading schools and adds each player's name.

Query 2 Results

The table below shows the career RBI leader from each of the top-ranked schools.

Rank	School	Top Player	Career RBI
1	Arizona State University	Barry Bonds	**1,996**
2	University of Southern California	Mark McGwire	**1,414**
3	University of California, Los Angeles	Todd Zeile	**1,110**
4	Stanford University	Bob Boone	**826**
5	University of Alabama	Joe Sewell	**1,055**
5	University of California, Berkeley	Jeff Kent	**1,518**
5	University of Miami	Mike Piazza	**1,335**
8	University of Michigan	George Sisler	**1,175**
9	University of Texas at Austin	Pinky Higgins	**1,075**
10	Georgia Institute of Technology	Mark Teixeira	**1,298**
10	Santa Clara University	Hal Chase	**941**

Arizona State's top producer is Barry Bonds, with 1,996 career RBI, the highest total of any school's leader. Other notable leaders include Jeff Kent of California–Berkeley (1,518), Mark McGwire of Southern California (1,414), and Mike Piazza of Miami (1,335). The list shows that the leading schools have produced players across many eras, from George Sisler at Michigan and Hal Chase at Santa Clara in the early 1900s to Mark Teixeira at Georgia Tech in recent decades.

Why This Is a Good Metric

1. **It follows management's own definition.** The metric applies the stated standard directly: a player is successful if he reaches 30 or more RBI. The threshold is clear, objective, and easy to repeat, and RBI is a standard statistic that baseball audiences already understand.
2. **It measures depth, not just one star.** Counting the number of different successful players, instead of adding up total RBI, rewards schools that consistently develop many productive hitters. Showing the percentage of each school's batters who succeeded also puts large and small programs on a fair footing.
3. **It is transparent and easy to audit.** The analysis is built in clear stages, and each stage can be reviewed or adjusted on its own. Window functions rank the schools consistently and handle ties openly, so another analyst could repeat the work and reach the same results.

Limitations of the Metric

1. **RBI depends on more than the batter.** A batter's RBI total depends heavily on how often teammates are on base ahead of him and where he bats in the lineup. A skilled hitter on a weak team can record fewer RBI than an average hitter on a strong one, so the metric does not measure individual skill alone.
2. **The 30 RBI threshold is a low bar.** A part-time regular can reach 30 RBI in a season, and the metric treats that player the same as a superstar. It shows which schools send productive hitters to the major leagues, but it does not separate good players from great ones.
3. **It credits attendance, not development.** The college records show where a player studied, not whether the school made him better. Players who attended more than one school are credited to each of them. The data also cannot see players who came from high school or from outside the college system, so it describes only part of where talent comes from. Schools with long baseball histories and large programs also have more chances to appear.

Additional Data That Would Improve the Metric

- **Draft and signing information.** Draft round, signing bonus, and whether a player was drafted out of college would show how highly professional teams rated each school's players, and would help separate a school's influence from a player's natural talent.
- **Advanced performance measures.** Measures such as Wins Above Replacement, on-base plus slugging, and league-adjusted statistics would give a fairer picture of quality than RBI, which depends on teammates.
- **Program context.** The number of years each player spent at a school, roster sizes, and program history would allow schools of different sizes and eras to be compared more fairly.

Conclusion

Using management's definition of 30 or more RBI, this analysis finds that Arizona State University produces the most successful players, followed by the University of Southern California and UCLA. UCLA and Miami stand out for turning the highest share of their major league batters into consistent run producers, while Texas shows that sending many players to the majors does not by itself mean a high success rate. The metric is a clear and defensible starting point, but it should be paired with measures of individual skill, draft information, and program context before management draws firm conclusions about which schools develop the best players.